

## Adding and Subtracting Fractions with Like Denominators

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*Adding, subtracting, and finding a missing numerator — all with like denominators*

**The big idea.** When two fractions have the **same denominator**, the pieces are the *same size*, so you can just count them.

- Add or subtract the **numerators** (the counts of pieces).
  - Keep the **denominator** (the size of one piece) exactly as it is.
  - Shade or cross out on the bar model first, then write the number sentence.
  - Write your final answer in simplest form when it can be simplified.
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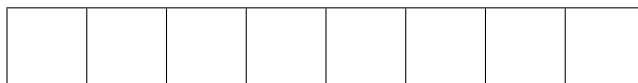
**1.** Each bar below is cut into **5** equal parts. Shade  $\frac{1}{5}$  on the first bar and  $\frac{3}{5}$  more on the same bar.



How much of the bar is shaded in all?  $\frac{1}{5} + \frac{3}{5} =$

Answer: \_\_\_\_\_

**2.** This bar is cut into **8** equal parts. Shade  $\frac{2}{8}$ , then shade  $\frac{5}{8}$  more.



Write the sum.  $\frac{2}{8} + \frac{5}{8} =$

Answer: \_\_\_\_\_

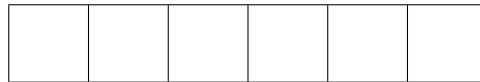
- 3.** Start by shading  $\frac{7}{9}$  of this bar, then cross out  $\frac{4}{9}$  of it.



How much is still shaded? Write the difference, then simplify it.  $\frac{7}{9} - \frac{4}{9} =$

Answer: \_\_\_\_\_

- 4.** A bar is cut into **6** equal parts and  $\frac{5}{6}$  of it is shaded. Cross out  $\frac{1}{6}$ .



Write the difference in simplest form.  $\frac{5}{6} - \frac{1}{6} =$

Answer: \_\_\_\_\_

5. Two bars, each cut into 8 equal parts, are shown side by side.



Shade  $\frac{5}{8}$  on the first bar and  $\frac{4}{8}$  on the second. Add them. The sum is bigger than one whole bar — write it as a fraction, and also say how many whole bars and how many eighths that is.

Answer: \_\_\_\_\_

6. **Missing numerator.** Fill in the blank so the sentence is true. Use the ninths bar to help you count up from  $\frac{2}{9}$  to  $\frac{7}{9}$ .



$$\frac{2}{9} + \underline{\hspace{2cm}} = \frac{7}{9}$$

Answer: \_\_\_\_\_

**7. Story problem.** A chocolate bar is cut into **12** equal pieces. Lena has  $\frac{11}{12}$  of the bar left over from yesterday. Today she eats  $\frac{5}{12}$  of the bar. What fraction of the bar is left now? Give the answer in simplest form.

Answer: \_\_\_\_\_

**8. Three parts of one whole.** Three friends shade parts of the same sixths bar: Ana shades  $\frac{1}{6}$ , Ben shades  $\frac{2}{6}$ , and Cruz shades  $\frac{2}{6}$ .



What fraction of the bar is shaded altogether? Is the bar completely full? Explain using the bar.

Answer: \_\_\_\_\_

**9. Work backwards.** A jug is marked in eighths. Someone poured out  $\frac{3}{8}$  of the jug, and  $\frac{4}{8}$  of the jug is still there.

What fraction of the jug was there before the pouring? Write the equation you used:

$$\text{_____} - \frac{3}{8} = \frac{4}{8}$$

Answer: \_\_\_\_\_

**10. Find and fix the mistake.** Jae wrote this on a test:

$$\frac{3}{7} + \frac{2}{7} = \frac{5}{14}$$

Use a sevenths bar to show why Jae's answer cannot be right, explain in one sentence what Jae did wrong, and write the correct sum.

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Answer: \_\_\_\_\_